

I2200 HW#2

Points: 100%

The images are attached as a zip file.

(1) Laplacian and Unsharp Masking (25%)

Show that subtracting the Laplacian from an image is proportional to unsharp masking. Use the definition for the Laplacian in the discrete case to derive the equivalence relation, constant weighting factors should not affect your conclusion.

(2) Spatial processing for photo enhancement (25%)

(2.1) Implement a function for histogram equalization and use it on "pout.png" below (do not use the `histeq()` function in MATLAB)

(2.1.1) [10%] Include your source code and the equalized image. Plot the intensity transformation function u vs. v obtained from the equalization function.

Compare the results with `histeq()` in matlab image processing toolbox.

(2.1.2) [5%] Experiment with contrast stretching with matlab or in an image editing software (e.g. GIMP, or one that you prefer) comment on the outcome in comparison with (2.1.1).

(2.2) [10%] Sharpen the input image leaf.jpg below, using a combinations of techniques in (2.1), and other matlab functions such as `fspecial`, `filter2`, etc. Compare your result with those from an image processing software, or the image leaf2.jpg.



pout.png



leaf.jpg



leaf (enhanced)

(3) High-frequency emphasis and histogram equalization (25%)

As illustrated in the following Figure, combining high-frequency emphasis and histogram equalization is an effective method for achieving edge sharpening and contrast enhancement.

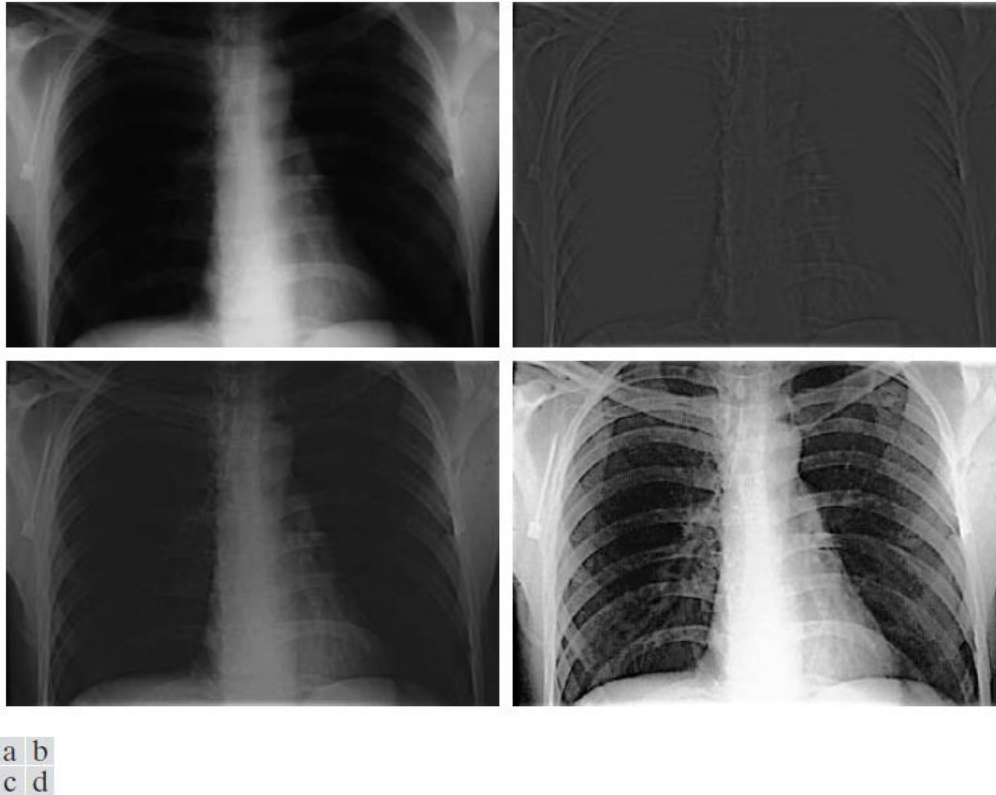


FIGURE 4.59 (a) A chest X-ray image. (b) Result of highpass filtering with a Gaussian filter. (c) Result of high-frequency-emphasis filtering using the same filter. (d) Result of performing histogram equalization on (c). (Original image courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School.)

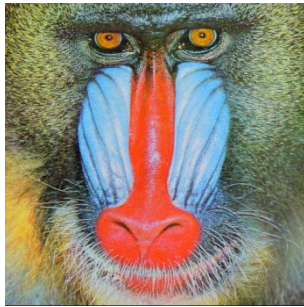
- (a) Show whether or not it matters which process is applied first.
- (b) If the order does matter, give a rationale for using one or the other method first.

(4) DFT and DCT on images (25%) – solve this question after Lecture 5

In this homework, we want to analyze the energy distributions of different types of images.

(4.1) [15%] Convert the input M-by-N color image to the grayscale format. Plot the 2-D log magnitude of the 2D DFT and DCT of the grayscale image, with center shifted. Visually compare and comment on the similarity/differences among the images using the two transforms.

(4.2) [10%] Apply the truncation windows discussed in the class to keep 25% and 6.25% ($1/4$ and $1/16$) of the DFT and DCT coefficients, i.e. two different ratios for each transform. This truncation is done by keeping the coefficients of the lowest frequencies (those within a centered smaller rectangle of $(M/2) \times (N/2)$ and $(M/4) \times (N/4)$ on the shifted FFT, respectively). Apply the 2D inverse DFT and inverse DCT to reconstruct the image for each of the truncated spectra. Compute the Signal-to-Noise-Ratio (SNR) value for each of the reconstructed images. Plot the reconstructed images visually examine and comment on the effects of truncation.



banboon



monkeyking**



sunflower*



hexagon

* from <http://www.flickr.com/photos/suneko/208994078/>, reused with the [creative commons license](#).

** from http://www.js.xinhuanet.com/zhuanlan/2005-05/19/content_4256139_3.htm, cartoon production by Shanghai Animation Studio 1961